
\$eq \$ne
\$lt \$gt
\$lte \$gte
\$in \$nin

COMPARISON OPERATORS

COMPARISION

The following operators can be used in queries to compare values:

- **\$eq**: Values are equal
 - **\$ne**: Values are not equal
 - **\$gt**: Value is greater than another value
 - **\$gte**: Value is greater than or equal to another value
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COMPARISION

- `$lt`: Value is less than another value
 - `$lte`: Value is less than or equal to another value
 - `$in`: Value is matched within an array
-

```
    age: 11,  
    Hobbies: [ 'Walking', 'Reading' ],  
    identity: { hasPanCard: false, hasAdhaarCard: true }  
  },  
  {  
    _id: ObjectId("6396a083f3b30b3e904d7951"),  
    name: 'Ghanshayam',  
    age: 11,  
    Hobbies: [ 'Walk', 'Cricket' ],  
    identity: { hasPanCard: false, hasAdhaarCard: true }  
  },  
  {  
    _id: ObjectId("6396a083f3b30b3e904d7952"),  
    name: 'Rita',  
    age: 11,  
    Hobbies: [ 'Anime' ],  
    identity: { hasPanCard: false, hasAdhaarCard: true }  
  },  
  {  
    _id: ObjectId("6396a083f3b30b3e904d7953"),  
    name: 'Ram',  
    age: 10,  
    Hobbies: [ 'Walk', 'Cricket' ],  
    identity: { hasPanCard: false, hasAdhaarCard: true }  
  },  
  {  
    _id: ObjectId("6396a083f3b30b3e904d7954"),  
    name: 'Geeta',  
    age: 12,  
    Hobbies: [ 'Gaming', 'Cooking' ],  
    identity: { hasPanCard: false, hasAdhaarCard: true }  
  }  
}
```

\$eq

- Specifies equality condition. The \$eq operator matches documents where the value of a field equals the specified value

Query: college> db.students.find({age:{seq:5}})

Example of \$eq

```
college> db.students.find({age:5})
[
  {
    _id: ObjectId("6396a083f3b30b3e904d794f"),
    name: 'Sita',
    age: 5,
    Hobbies: [ 'Walk', 'Cricket' ],
    identity: { hasPanCard: false, hasAdhaarCard: true }
  }
]
college> db.students.find({age:{$eq:5}})
[
  {
    _id: ObjectId("6396a083f3b30b3e904d794f"),
    name: 'Sita',
    age: 5,
    Hobbies: [ 'Walk', 'Cricket' ],
    identity: { hasPanCard: false, hasAdhaarCard: true }
  }
]
college> |
```

\$gt

- \$gt selects those documents where the value of the specified field is greater than (i.e. >) the specified value
- The \$gt operator has the following form:

{ field: { \$gt: value } }

Example of \$gt

By using gt method

college> db.student.find{age:{gt:5}}

Query: college> db.students.find({age:{\$gte:5}})

Result:

```
{
  _id: ObjectId("6396a083f3b30b3e904d7950"),
  name: 'Shyam',
  age: 11,
  Hobbies: [ 'Walking', 'Reading' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
{
  _id: ObjectId("6396a083f3b30b3e904d7951"),
  name: 'Ghanshayam',
  age: 11,
  Hobbies: [ 'Walk', 'Cricket' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
{
  _id: ObjectId("6396a083f3b30b3e904d7952"),
  name: 'Rita',
  age: 11,
  Hobbies: [ 'Anime' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
{
  _id: ObjectId("6396a083f3b30b3e904d7953"),
  name: 'Ram',
  age: 10,
  Hobbies: [ 'Walk', 'Cricket' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
```

\$gte

- \$gte selects the documents where the value of the specified field is greater than or equal to (i.e. $\geq$) a specified value (e.g. `value`.)
 - **\$gte**: Value is greater than or equal to another value
 - Result:
-

```
    age: 11,  
    Hobbies: [ 'Walking', 'Reading' ],  
    identity: { hasPanCard: false, hasAdhaarCard: true }  
  },  
  {  
    _id: ObjectId("6396a083f3b30b3e904d7951"),  
    name: 'Ghanshayam',  
    age: 11,  
    Hobbies: [ 'Walk', 'Cricket' ],  
    identity: { hasPanCard: false, hasAdhaarCard: true }  
  },  
  {  
    _id: ObjectId("6396a083f3b30b3e904d7952"),  
    name: 'Rita',  
    age: 11,  
    Hobbies: [ 'Anime' ],  
    identity: { hasPanCard: false, hasAdhaarCard: true }  
  },  
  {  
    _id: ObjectId("6396a083f3b30b3e904d7953"),  
    name: 'Ram',  
    age: 10,  
    Hobbies: [ 'Walk', 'Cricket' ],  
    identity: { hasPanCard: false, hasAdhaarCard: true }  
  },  
  {  
    _id: ObjectId("6396a083f3b30b3e904d7954"),  
    name: 'Geeta',  
    age: 12,  
    Hobbies: [ 'Gaming', 'Cooking' ],  
    identity: { hasPanCard: false, hasAdhaarCard: true }  
  }  
]
```

\$in

- The \$in operator selects the documents where the value of a field equals any value in the specified array.
 - **\$in**: Value is matched within an array
 - using

```
college> db.students.find({age:{$in:[5,12]}})
```
 - Result:
-

```
{
  _id: ObjectId("6396a083f3b30b3e904d794f"),
  name: 'Sita',
  age: 5,
  Hobbies: [ 'Walk', 'Cricket' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
},
{
  _id: ObjectId("6396a083f3b30b3e904d7954"),
  name: 'Geeta',
  age: 12,
  Hobbies: [ 'Gaming', 'Cooking' ],
  identity: { hasPanCard: false, hasAdhaarCard: true }
}
```

\$nin

\$nin selects the documents where:

- the specified field value is not in the specified array **or**
 - the specified field does not exist.
-

```
college> db.students.find({age:{$nin:[5,12]}})
[
  {
    _id: ObjectId("6396a083f3b30b3e904d7950"),
    name: 'Shyam',
    age: 11,
    Hobbies: [ 'Walking', 'Reading' ],
    identity: { hasPanCard: false, hasAdhaarCard: true }
  },
  {
    _id: ObjectId("6396a083f3b30b3e904d7951"),
    name: 'Ghanshayam',
    age: 11,
    Hobbies: [ 'Walk', 'Cricket' ],
    identity: { hasPanCard: false, hasAdhaarCard: true }
  },
  {
    _id: ObjectId("6396a083f3b30b3e904d7952"),
    name: 'Rita',
    age: 11,
    Hobbies: [ 'Anime' ],
    identity: { hasPanCard: false, hasAdhaarCard: true }
  },
  {
    _id: ObjectId("6396a083f3b30b3e904d7953"),
    name: 'Ram',
    age: 10,
    Hobbies: [ 'Walk', 'Cricket' ],
    identity: { hasPanCard: false, hasAdhaarCard: true }
  }
]
```

Example:

```
college> db.students.find({ Hobbies: 'Walk' })
[
  {
    _id: ObjectId("6396a083f3b30b3e904d794f"),
    name: 'Sita',
    age: 5,
    Hobbies: [ 'Walk', 'Cricket' ],
    identity: { hasPanCard: false, hasAdhaarCard: true }
  },
  {
    _id: ObjectId("6396a083f3b30b3e904d7951"),
    name: 'Ghanshayam',
    age: 11,
    Hobbies: [ 'Walk', 'Cricket' ],
    identity: { hasPanCard: false, hasAdhaarCard: true }
  },
  {
    _id: ObjectId("6396a083f3b30b3e904d7953"),
    name: 'Ram',
    age: 10,
    Hobbies: [ 'Walk', 'Cricket' ],
    identity: { hasPanCard: false, hasAdhaarCard: true }
  }
]
college> |
```

Example:

```
[
  {
    _id: ObjectId("6396a083f3b30b3e904d7953"),
    name: 'Ram',
    age: 10,
    Hobbies: [ 'Walk', 'Cricket' ],
    identity: { hasPanCard: false, hasAdhaarCard: true }
  }
]
```

\$lt

Syntax: { field: { \$lt: value } }

[\\$lt](#) selects the documents where the value of the field is less than (i.e. <) the specified value.

For most data types, [comparison operators](#) only perform comparisons on fields where the [BSON type](#) matches the query value's type. MongoDB supports limited cross-BSON comparison through [Type Bracketing](#).

\$lte

Syntax: { field: { \$lte: value } }

[\\$lte](#) selects the documents where the value of the field is less than or equal to (i.e. $\leq$) the specified value.

For most data types, [comparison operators](#) only perform comparisons on fields where the [BSON type](#) matches the query value's type. MongoDB supports limited cross-BSON comparison through [Type Bracketing](#).

Example

```
db.inventory.insertMany( [  
  {  
    "item": "nuts", "quantity": 30,  
    "carrier": { "name": "Shipit", "fee": 3 }  
  },  
  {  
    "item": "bolts", "quantity": 50,  
    "carrier": { "name": "Shipit", "fee": 4 }  
  },  
]
```

Example

```
{  
  "item": "washers", "quantity": 10,  
  "carrier": { "name": "Shipit", "fee": 1 }  
}  
]
```

Query: db.inventory.find({ quantity: { \$lte: 20 } })

Example output

```
{  
  
  _id: ObjectId("61ba453ffe687fce2f04241c"),  
  
  item: 'washers',  
  
  quantity: 10,  
  
  carrier: { name: 'Shipit', fee: 1 }  
  
}
```